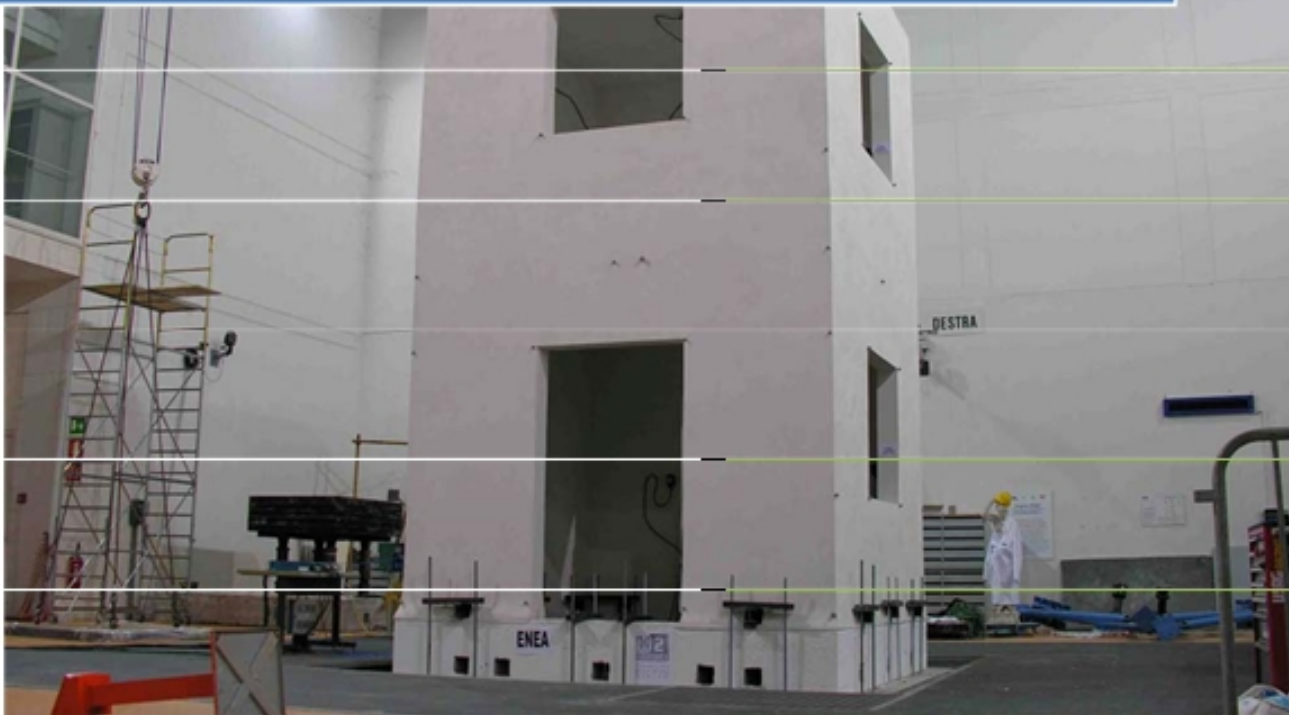


ENEA – EMMEDUE Agreement
Inspection and validation of buildings
constructed with the EMMEDUE system
through experimental static and seismic
behavioral tests.

SEISMIC ANALYSES OF EMMEDUE SUBSYSTEMS UPON VIBRATING TABLE: ACCELEROMETRIC READINGS



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ENEA - ACS

February 2009

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1 INTRODUCTION

This report shall be included within the agreement stipulated by EMMERDUE and ENEA for the “Inspection and validation of buildings constructed with the EMMEDUE system through experimental static and seismic behavioral tests”, dated 10/10/2007, which placed ENEA in charge of the following activities:

- Dynamic testing of 2 subsystems upon vibrating tables,
- Evaluation and guidelines,
- Coordination of the entire project.

The experimental phase upon the vibrating table is actually made up of the testing of 3 subsystems:

- 1) A C-shape designed sub system, made up of a 2.70 m high and 3.00 m long wall, with two 1.00 m long wall portions at the extremities and a floor at the top;
- 2) An H-shape designed sub system, made up of a 2.70 m high and 3.00 m long wall, with two 1.00 m long wall portions at the extremities and a floor at the top;
- 3) A sub system made up of four perimeter walls, of which, on the lower floor level, two possess door openings, one includes a window opening and one does not have any openings. All of the walls have window openings on the upper floor; the walls have a design length of 3.40 m (excluding the curb) and a total height of 5.74m (excluding the curb), with a floor on top, as well as at an intermediate height, in order to simulate a two-level building portion of equal or approximate scale to that of the real portion.

The samples were all built by EMMEDUE at the C.R. Casaccia Dynamic Testing Laboratory from 14/07 to 08/08/2008. The tests were performed on the dates 24/10/2008 (case 1), 30/10/2008 (case 2) and from 25 to 27/11/2008 (case 3).

2 TWO-LEVEL BUILDING MODEL

2.1 Description of the model

The two level model has a square design shape, 340 cm per side, total height equal to 574 cm and a distance between floors of 287 cm. On the first floor the N and S walls both include a door opening, the E wall has a window opening and the W wall has no openings; on the second floor each wall has a window opening. The walls are supported by a 25x50 cm reinforced cement curb which has been anchored to the table with metallic profiling.

The walls, which have a total thickness of 15cm, are made up of a prefabricated core, made from an 8 cm polystyrene panel with two 3.5mm and 2.5mm reinforcement mesh nets welded vertically and horizontally, respectively, and connected by about 70 metallic connectors per m² and two spritz beton applied wall dressings on each face,

each of a thickness equal to 3.5 cm, to complete the wall. The floors are made in the same manner with 17cm thick panels, an 8cm polystyrene core, a 5.5cm upper floor cast on site and a 3.5cm spritz beton lower floor.

The weight of the walls is 1.60 kN/m^2 and the floor weighs 2.10 kN/m^2 . Bags of cement have been positioned upon the first level covering in order to simulate a load of about 4.5 kN/m^2 . No load, on the other hand, was applied to the covering floor. This choice was determined by the necessity to keep the total center of gravity as low as possible and, therefore, to be able to apply increased seismic accelerations at the bas of the model.

The model, built by EMMEDUE at the C.R. Casaccia Dynamic Testing Laboratory (Figure 1) from 14/07 to 08/08/2008, was subjected to finishing operations, again by EMMEDUE, essentially consisting in the application of stucco, from 10 to 15/11/2008. On 20/11/2008 the model was loaded upon the vibrating table for testing (Figure 2, Figure 3). The tests were performed over the subsequent days, from 24/11 to 25/11, and were completed on 27/11 with a day of study (Figure 4).



Figure 1 The model during its construction



a)



b)



c)

Figure 2 Loading of the model upon the vibrating table



Figure 3 Fastening of the model to the vibrating table



Figure 4 The model during the tests

2.2 Instrumentation and tests

The model's vibrations were measured with 2 instrumentation systems:

- 18 FBA11 Accelerometers connected to 2 K2 Kinemetrics detectors, each with an internal accelerometric triad, for a total of 24 sensors;
- 3D VISION system and 4 additional accelerometric triads.

This technical report provides the results relative to the first instrumentation system. The results relative to the second instrumentation system shall be provided in another technical report. The structure was stimulated at its base with the accelerograms recorded in the two directions at the following sites and during the following seismic events:

- Colfiorito, Umbria-Marche Earthquake, 1997
- Tolmezzo, Friuli 1976.

The accelerograms were scaled in such a way to as to achieve a variable peak from 0.05g to 0.45g, at intervals of 0.05g. The maximum value of the accelerations was determined by the table's capacity. Each seismic series was separated by dynamic characterization testing in both directions, with sine sweep and white noise. A synthetic accelerogram was even used during the final testing phase. Table 1 lists the complete testing sequence.

Table 1 Seismic testing sequence

Test	Accelerogram	Direction	File AQ1	File AQ2
WN01X	White Noise	X	LZ004	PD004
WN01Y	White Noise	Y	LZ009	PD010
SS01X	Sine Sweep	X	LZ010	PD011
SS01Y	Sine Sweep	Y	LZ012	PD013
CO005	Colfiorito	XY	LZ013	PD014
TO005	Tolmezzo	XY	LZ014	PD015
WN02X	White Noise	X	LZ015	PD016
WN02Y	White Noise	Y	LZ016	PD017
CO010	Colfiorito	XY	LZ017	PD018
TO010	Tolmezzo	XY	LZ018	PD019
WN03X	White Noise	X	MA004	PE004
WN03Y	White Noise	Y	MA005	PE005
SS03X	Sine Sweep	X	MA006	PE006
SS03Y	Sine Sweep	Y	MA008	PE008
CO015	Colfiorito	XY	MA009	PE009
TO015	Tolmezzo	XY	MA010	PE010
WN04X	White Noise	X	MA011	PE011
WN04Y	White Noise	Y	MA012	PE012
CO020	Colfiorito	XY	MA013	PE013
TO020	Tolmezzo	XY	MA014	PE014
WN05X	White Noise	X	MA015	PE015
WN05Y	White Noise	Y	MA016	PE016
SS05X	Sine Sweep	X	MA017	PE017
SS05Y	Sine Sweep	Y	MA018	PE018
WNb05X	White Noise	X	MA019	PE019
WNb05Y	White Noise	Y	MA020	PE020
CO025	Colfiorito	XY	MA021	PE021
TO025	Tolmezzo	XY	MA022	PE022

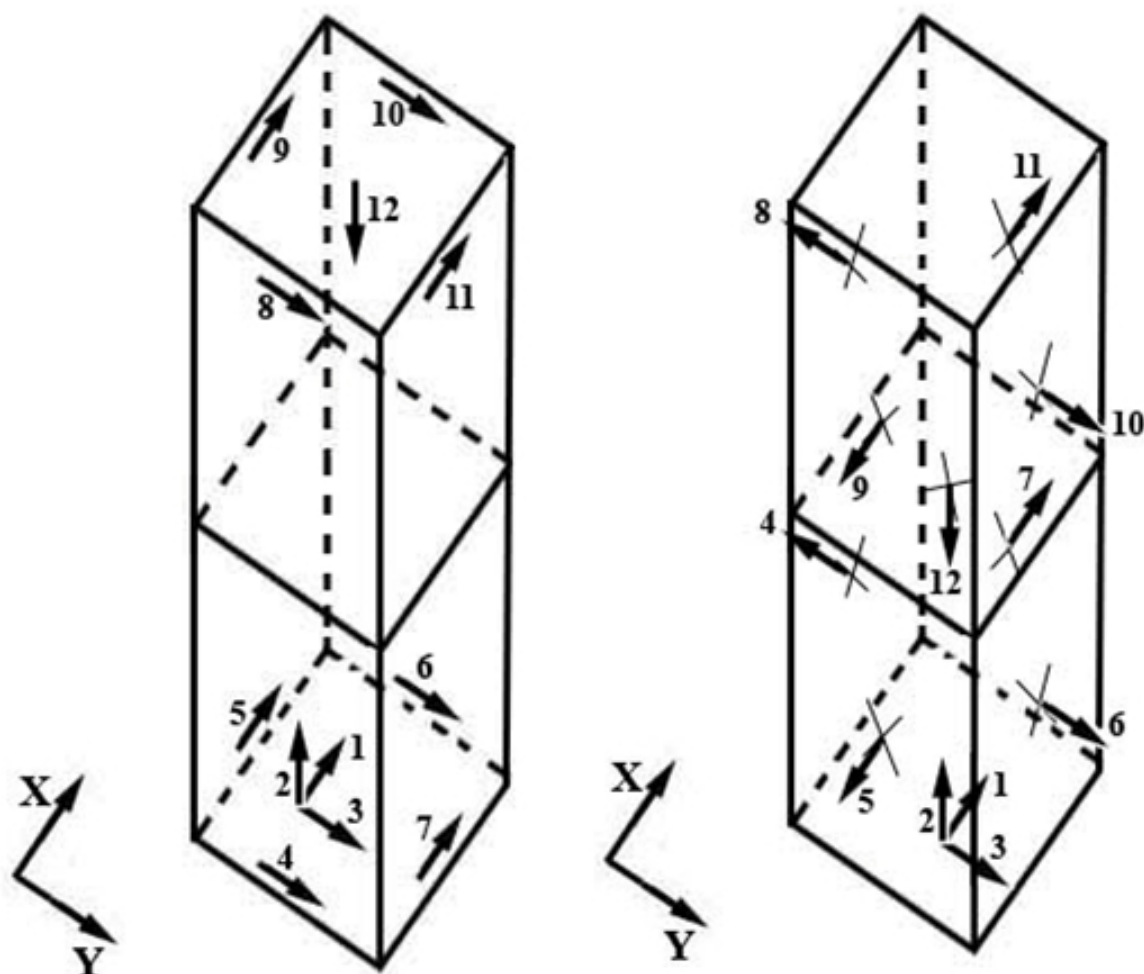
WN06X	White Noise	X	MA023	PE023
WN06Y	White Noise	Y	MA024	PE024
CO030	Colfiorito	XY	MB002	PF002
TO030	Tolmezzo	XY	MB003	PF003
WN07X	White Noise	X	MB004	PF004
WN07Y	White Noise	Y	MB005	PF005
CO035	Colfiorito	XY	MB006	PF006
TO035	Tolmezzo	XY	MB007	PF007
WN08X	White Noise	X	MB008	PF008
WN08Y	White Noise	Y	MB009	PF009
CO040	Colfiorito	XY	MB010	PF010
TO040	Tolmezzo	XY	MB011	PF011
WN09X	White Noise	X	MB013	PF013
WN09Y	White Noise	Y	MB014	PF014
CO045	Colfiorito	XY	MB015	PF015
TO045	Tolmezzo	XY	MC001	PG001
WN10X	White Noise	X	MC002	PG002
WN10Y	White Noise	Y	MC003	PG003
SI020	Synthetic	XY	MC004	PG004
SI030	Synthetic	XY	MC005	PG005
SI040	Synthetic	XY	MC006	PG006
SI045	Synthetic	XY	MC007	PG007
WN11X	White Noise	X	MC008	PG008
WN11Y	White Noise	Y	MC009	PG009

As previously stated, the model was monitored with two 12-channel K2 Kinematics detectors, 3 internal and 9 external connected to the same number of FBA11 Kinematics accelerometers. Each detector also contains a PCMCIA card, a battery and a triad of FBA accelerometric sensors and has a scale base of ± 2.5 Volt. The signals continuously detected by the sensors are normally analyzed by the K2 in order to verify whether the assigned dynamic conditions have been exceeded (surpassing of a threshold value in one or more sensors, etc.) If, and only if, this should occur, the system saves the data to the PCMCIA card from which they can be read by a PC. In the case at hand, the saving of the data was commanded manually. The sampling frequency was set to 200 samples/sec.

The two detectors were fastened directly to the vibrating table; in this manner the two internal triads (sensors 101-103 of AQ1 and sensors 201-203 of AQ2) measured the input at the base of the structure. The other sensors, on the other hand, were arranged upon the structure according to the following scheme:

- Sensors 104 to 112, connected to the AQ1 detector, were positioned in the following manner: 4 upon the base curb in a horizontal direction parallel to the level of each of the walls, 4 upon the covering in correspondence to those at the base and one sensor at the center of the covering floor in a vertical direction;
- Sensors 204 to 212, connected to the AQ2 detector, were positioned in the following manner: 8 upon each of the walls in a horizontal direction, orthogonal to the level of each of the walls themselves; more specifically, the sensors were positioned upon 7 walls in relation to the borders of the openings present (both the first and second levels) and at the center of the wall in the only case where no openings were present; finally, one sensor was positioned at the center of the first floor in a vertical direction.

From the readings of the sensors connected to the AQ1 detector it is possible to recognize the general inflexion methods, while from those connected to the AQ2 detector it is possible to highlight even the local methods outside the individual floor of the single walls. The scheme of the sensors is shown in Figure 5, while Figure 6 shows some photographs of the instrumentation.



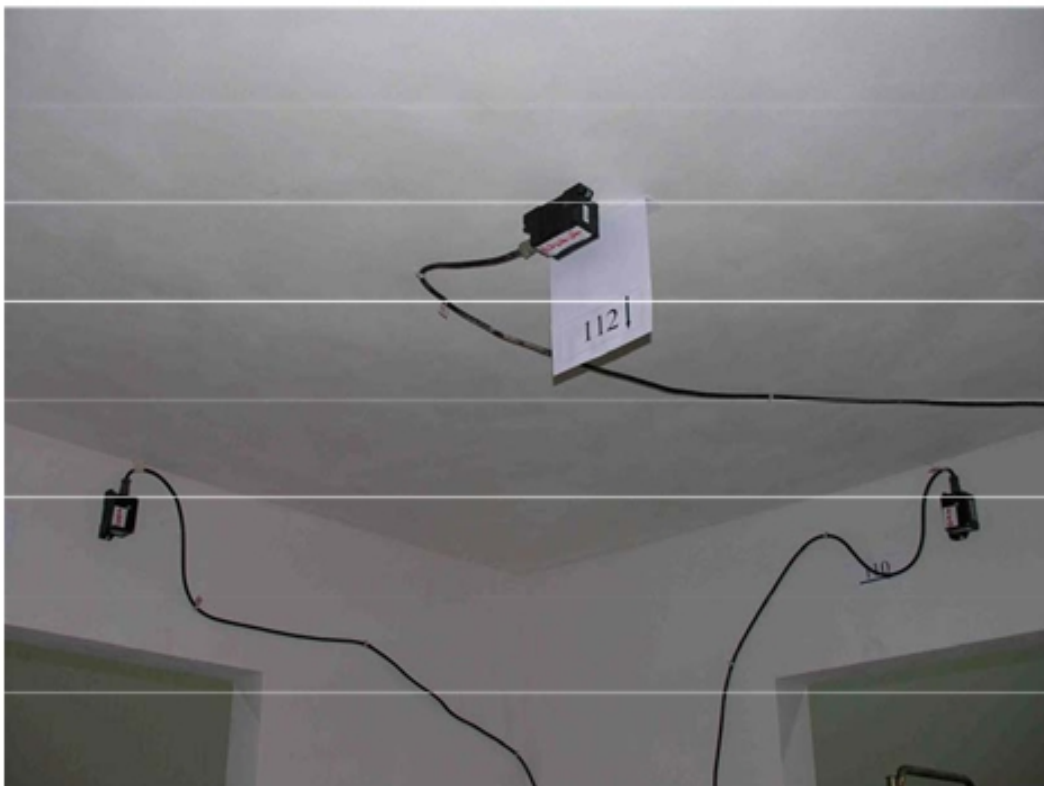
Sensors connected to the AQ1 detector

Sensors connected to the AQ2 detector

Figure 5 Sensor positions upon the model



a)



b)

Figure 6 a) K2 detectors and b) sensors 110, 111 and 112

2.3 Data analyses

All of the readings were analyzed:

- in the field of time (th);
- in the field of frequency, illustrating: The power spectral auto-densities (PSD); the cross spectral power densities (CSD), for all of the pairs of sensors, in terms of amplitude and phase factors with the relative coherence functions; the transfer functions from the base to the summit for the most significant sensors.

Please be reminded that the presence of peaks in the PSD graphics indicates possible structural resonance frequencies; the confirmation of this is shown by the presence of peaks in the cross spectra in relation to the same frequencies with significant phase factor values and coherence function values. Peaks which do not correspond to structural resonance can be caused by the nature of the input or other factors. Keeping in mind the phase factors, information regarding the modal forms can be collected from the spectral amplitudes.

Downstream from the present report can be found the graphics (time-history, auto-spectra, cross-spectra and transfer functions) regarding characterization and seismic testing:

- CO045 (Colfiorito. PGA = 0.45g)
- TO045 (Tolmezzo. PGA = 0.45g)
- SI045 (Synthetic. PGA = 0.45g)

During the entire testing cycle no damage was encountered, even in relation to the highest levels of applied stress. The analysis of the graphics produced confirmed the substantial preservation of the model's dynamic characteristics.

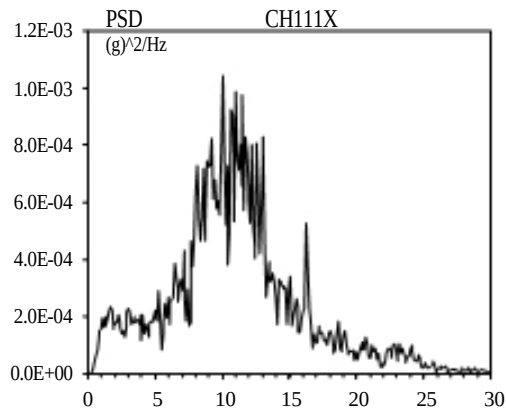
2.3.1 Dynamic characterization testing

The analysis of the time-history did not reveal a significant amplification at the summit with respect to the base, indicating considerable model rigidity.

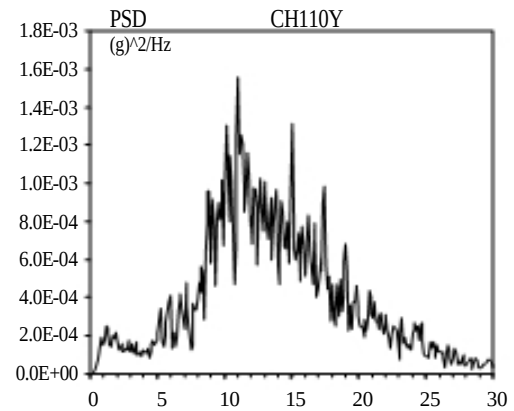
Figure 7 shows the auto-spectra relative to the initial dynamic characterization performed before subjecting the model to the seismic tests. Evident natural structural frequencies, around 10 Hz, appear for both directions. It must be noted, however, that due to the model's high rigidity, the auto-spectra do not clearly highlight the amplification in relation to the natural frequencies. Analogous results can be seen in all of the other characterizations.

In reference to another characterization test (Figure 8), the transfer functions clearly highlight the first structural frequency of 10Hz in the Y direction. In the X direction, the first structural frequency is between 11 and 12 Hz (Figure 9), which is coherent with the increased rigidity of the structure in this direction.

As indicated above, no variations were observed in the structural frequency identified for the different characterizations, which confirms that no type of structural damage took place; for this purpose, as an example, analogous diagrams are shown for the final characterization in the Y direction (Figure 10).

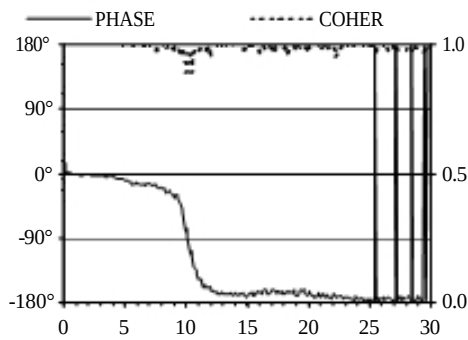
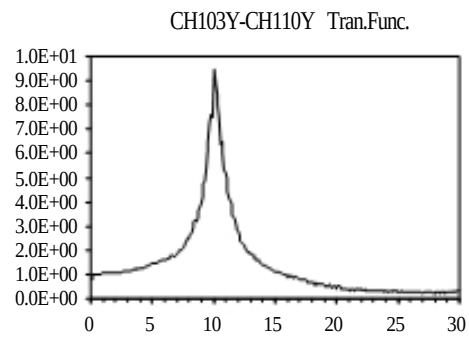
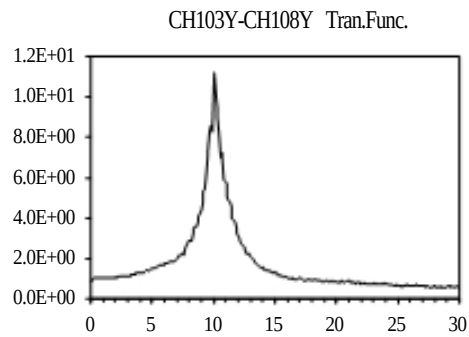


PSD Initial X Characterization AQ1

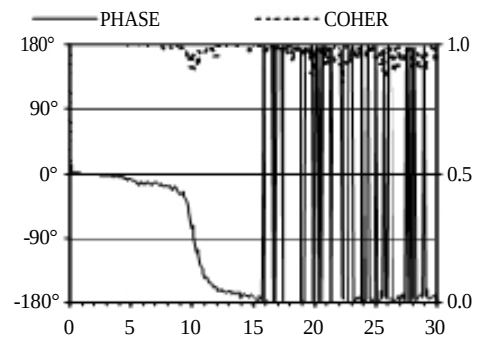


PSD Initial Y Characterization AQ1

Figure 7



Tr.Fun. CH08 Charac. WN06Y



Tr.Fun. CH10 Charac. WN06Y

Figure 8

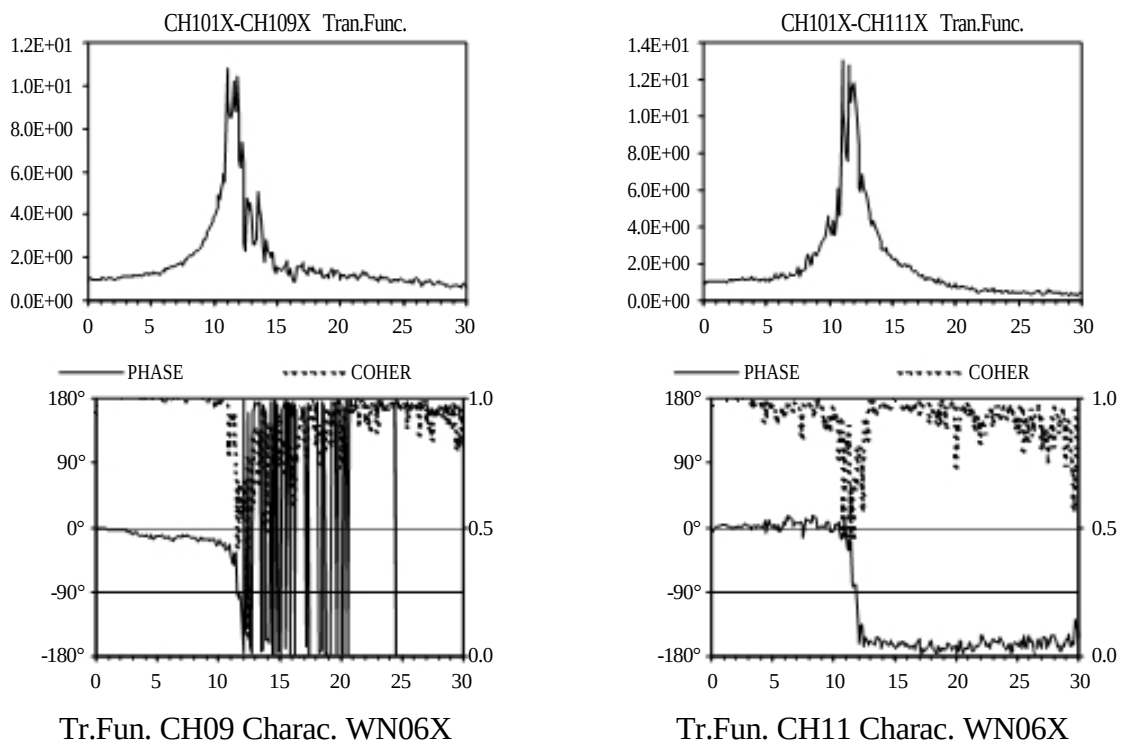


Figure 9

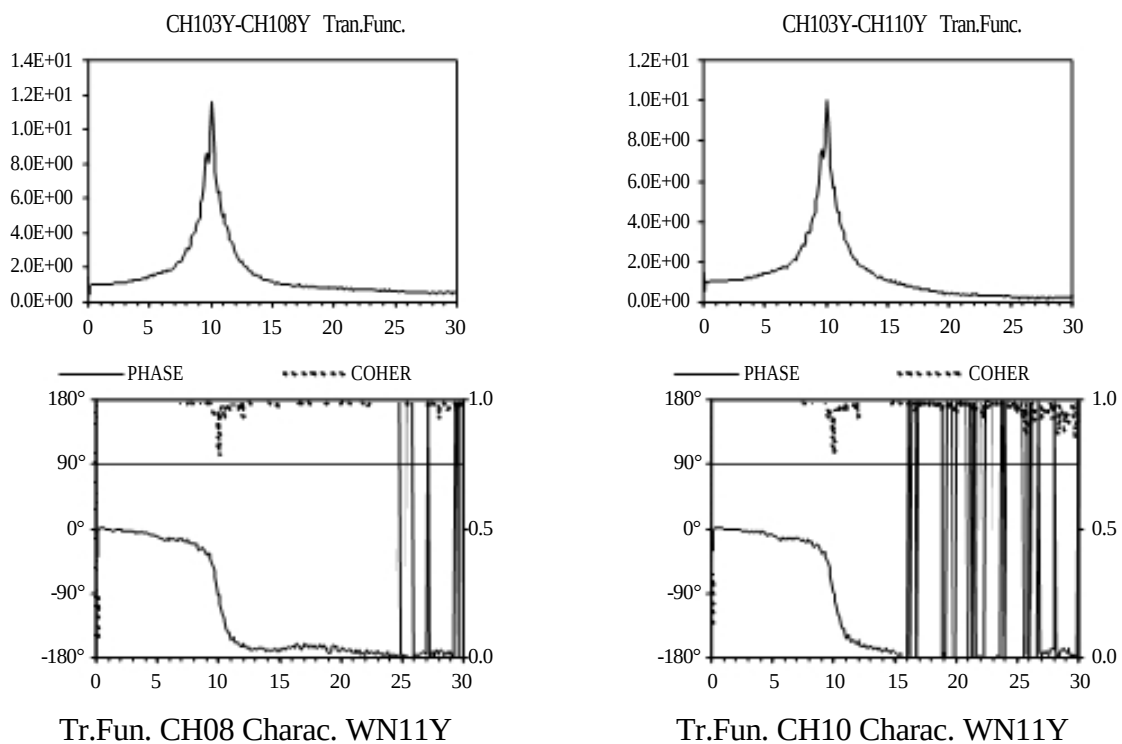


Figure 10

2.3.2 Seismic tests

Even in the seismic tests, the analysis of the time-history showed no significant amplification at the summit with respect to the base while in the auto-spectra the amplification in relation to the natural frequencies was not clear. In fact, the input frequency content is present upon the entire structure, masking the natural dynamic characteristics. In Figure 11, relative to the 0.25g Colfiorito case, it can be observed that the input crosses the structure without significant modifications.

Nevertheless, various peaks between 10 and 12 Hz are evident.

Moreover, a slight reduction of the natural frequencies is noted. This results both from the increased stress level, which may trigger non-linear structural behavior, as well as from the presence of nearby frequencies caused by the input, which blend in with the structural frequencies; in particular, in Figure 12, it can be noted how the X direction frequency drops to about 10 Hz while that of the Y direction drops to about 9 Hz.

As indicated above, no frequencies were identified which make reference to inflexions outside of the floor of the walls due to the proximity of the sensors positioned upon the borders of the openings; the auto-spectrum relative to sensor 204, however, which was located at the center of the only wall without openings, revealed a peak of 31.7 Hz (Figure 13). This data can help to determine useful indications for modeling the flexional rigidity of the walls.

That which has been affirmed for the auto-spectra, regarding the difficulties in identifying the natural frequencies, also holds true for the cross-spectra, which do not provide certain values for the structural frequencies; rather, frequencies which are also seen in the base sensors are often noted and can be clearly ascribed to the input (Figure 14). A useful indication which has been provided arises from the fact that the crosses between parallel sensors located upon the same level have a null phase, and therefore there is no torsion; the modal form is of a flexional typology.

The sensors arranged in a vertical direction show the following frequencies: 25.2 Hz for the covering floor and about 15.8 Hz for the intermediate floor (Figure 15), values which are coherent with the different mass present upon each floor. Like for that of the walls, this data can help to determine useful indications for modeling the flexional rigidity of the floors.

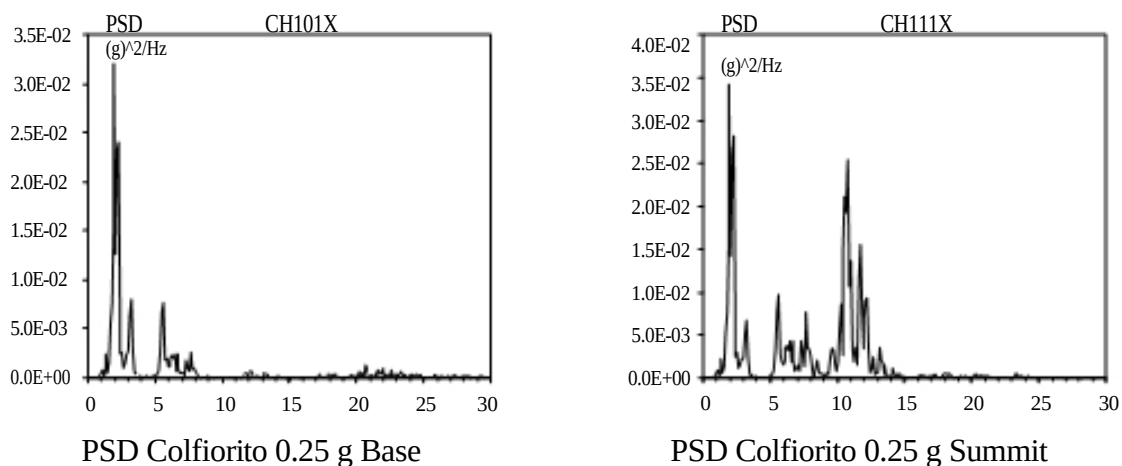


Figure 11

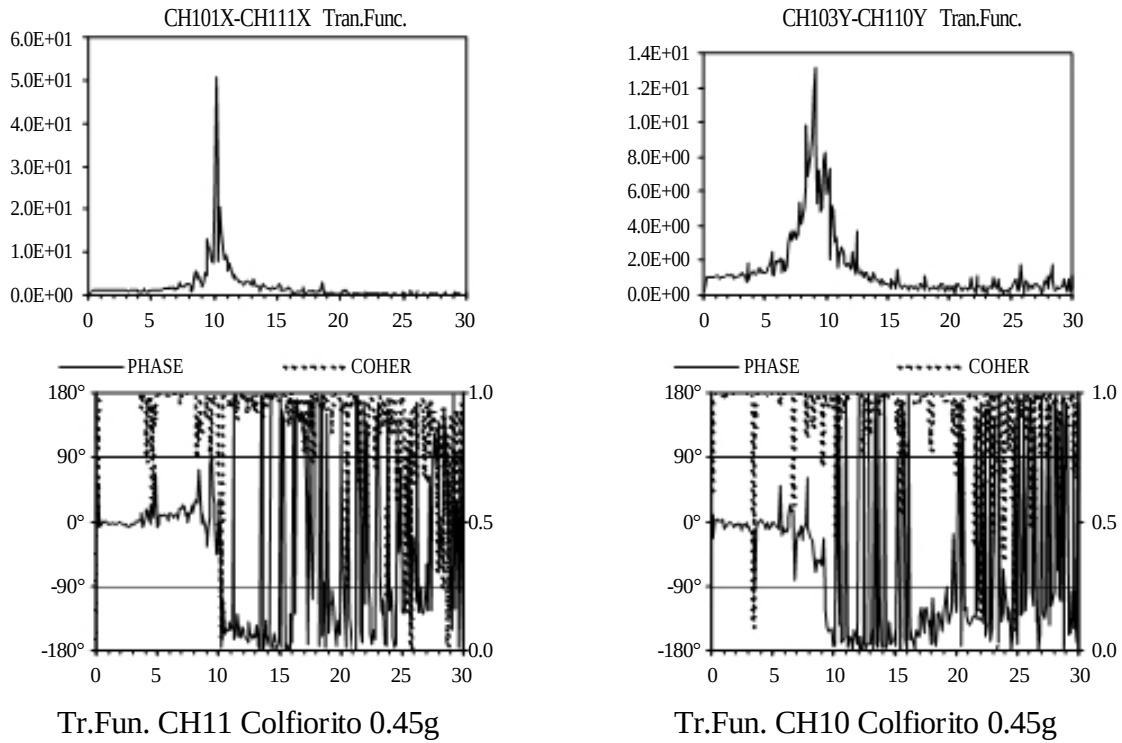


Figure 12

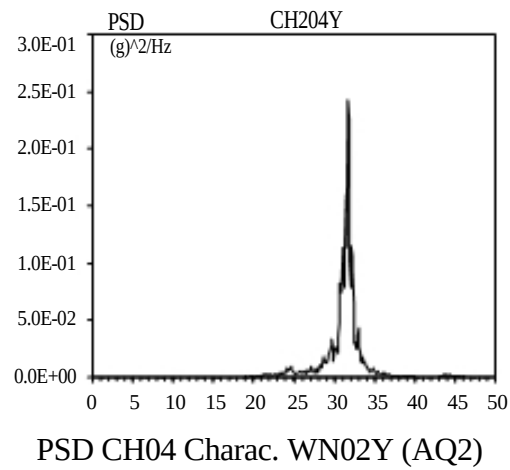


Figure 13

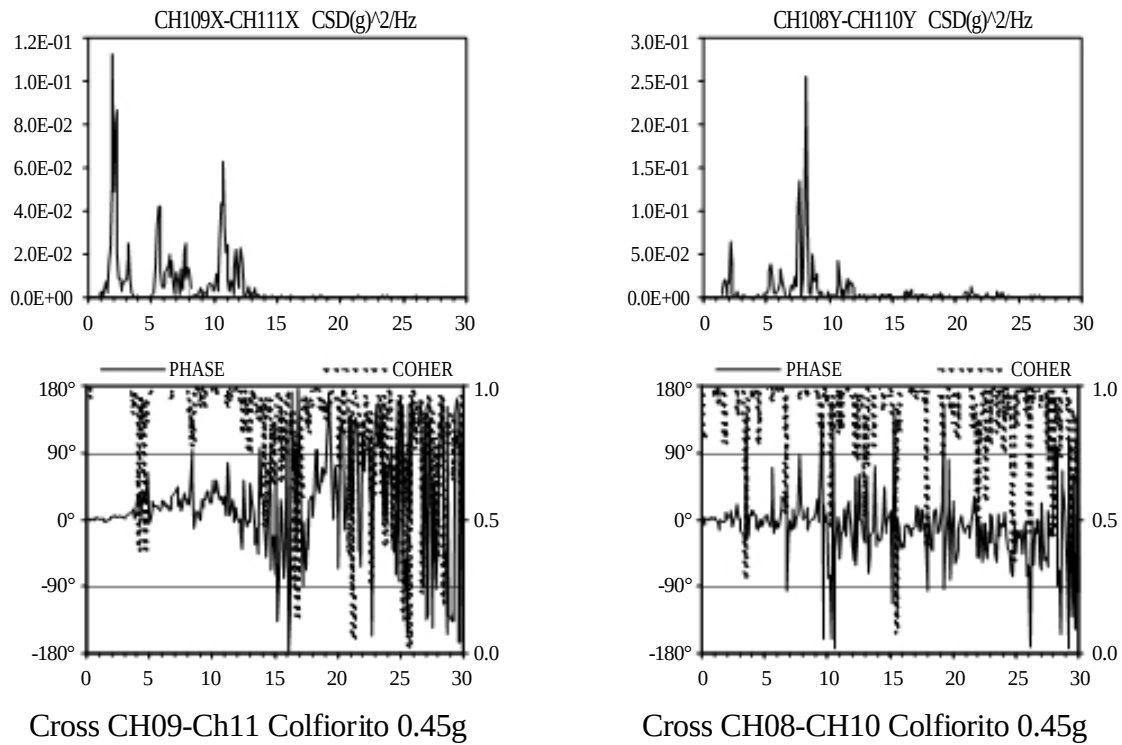


Figure 14

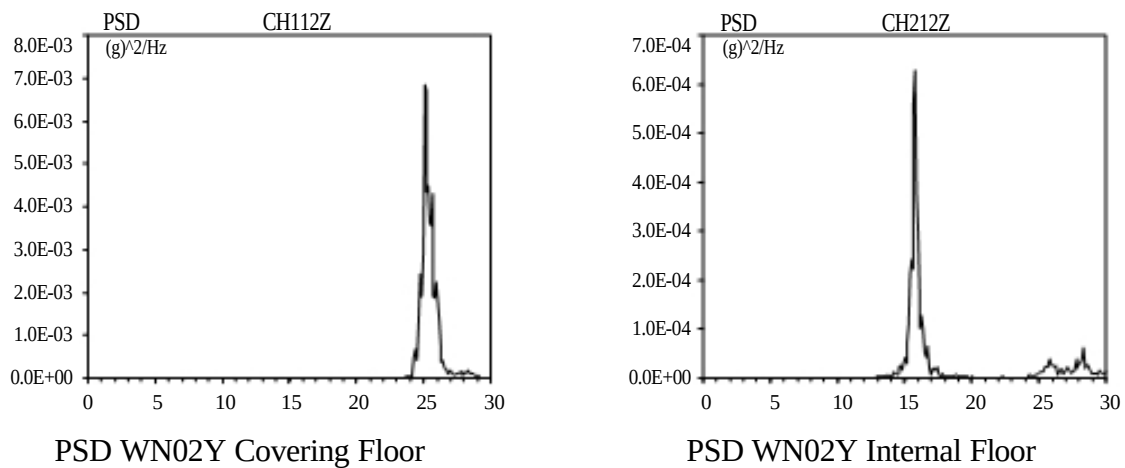


Figure 15

3 PRELIMINARY SUBSYSTEMS TESTS

3.1 Model C

The C shaped Model was set up using piezoelectric sensors which took readings at sampling intervals of 100 points/sec. The sensors were positioned upon the table and at 4 levels along the model according to X-Y pairs arranged along one of the uprights;

An additional pair was positioned upon another upright and an X direction sensor was placed at the center of the core. Figure 16 indicates the measurement points and their denominations; the X direction is that which is perpendicular to the core of the C, while the Y direction is that which is parallel to it.

The tests upon this model were performed using two real accelerograms, registered respectively at Colfiorito (1997) and at Tolmezzo (1976), as well as a synthetic accelerogram; the stress level was pushed to a peak of 0.8g. The seismic tests were alternated with dynamic characterization tests, performed by means of sine-sweep. From the initial characterization (Fig. 17), a frequency of 8.5 Hz can be clearly recognized in the Y direction, while a less evident peak of slightly more than 10.1 Hz can be seen in the X direction. It must be noted that the model should be more rigid in the Y direction; this paradox probably results for an inefficient fastening to the vibration table.

The sensor at the center of the core (Fig. 18) shows a frequency of 29.0 Hz, which is compatible with the bulging modal form of the C model's core.

The seismic tests highlight peaks at lower levels with respect to those which were first shown for the characterizations, which were linked to the input characteristics.

Moreover, the scarce efficacy of the attachment to the base became more evident as the stresses increased. The model, save for its attachment to the vibration table, passed the testing without suffering evident damage.

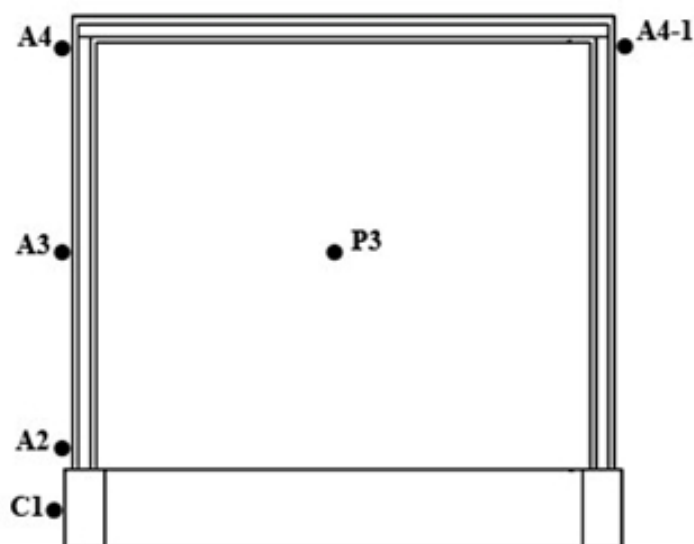


Figure 16

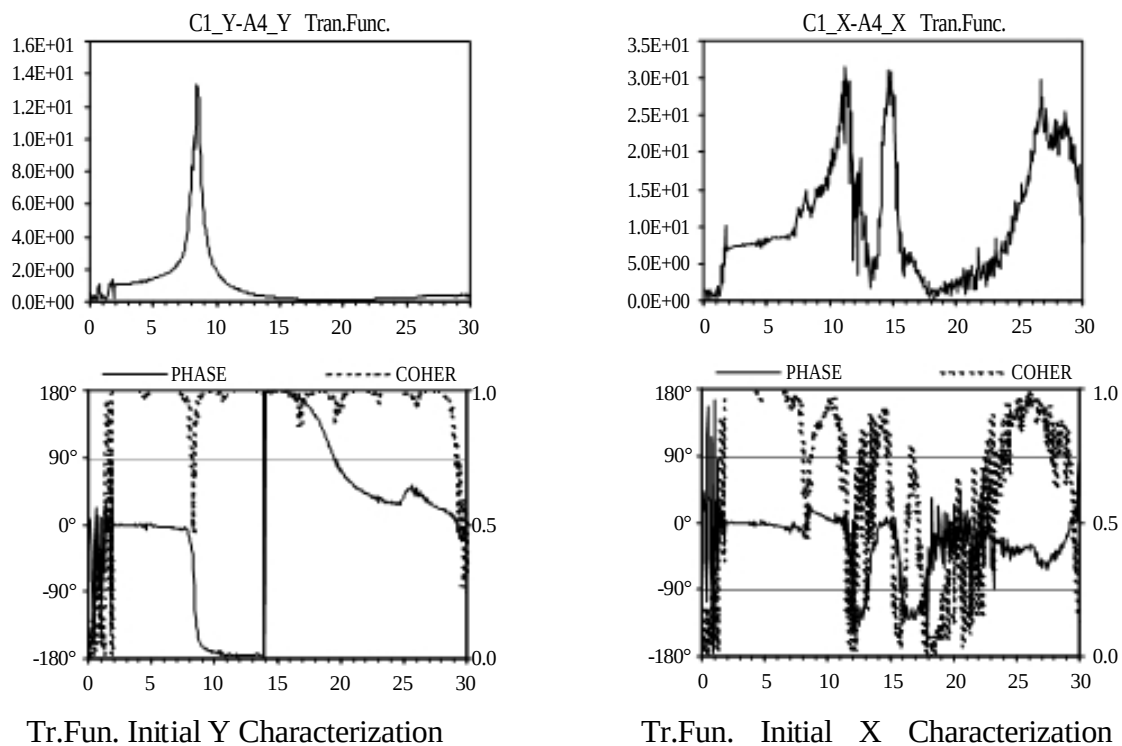


Figure 17

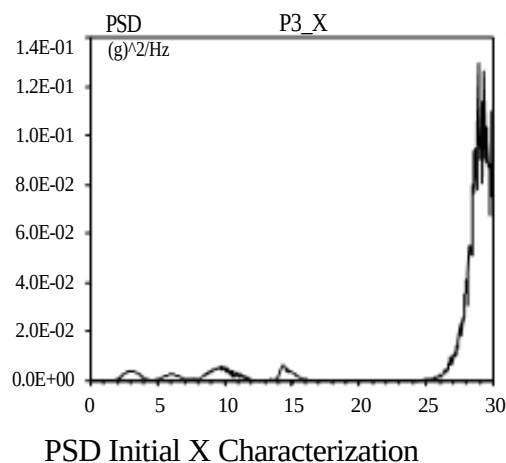


Fig. 18

3.2 Model H

The arrangement of the instruments is exactly like that of model C (Fig. 16). The seismic tests in this case reached a peak of 1.0 g. The results (Fig. 19) highlighted an increased rigidity in the Y direction with two frequencies at about 14 Hz and at about 23 Hz. The behavior in the X direction is less clear as an amplification zone between 5 and 10 Hz can be noted. This model also passed the seismic testing without suffering any evident damage.

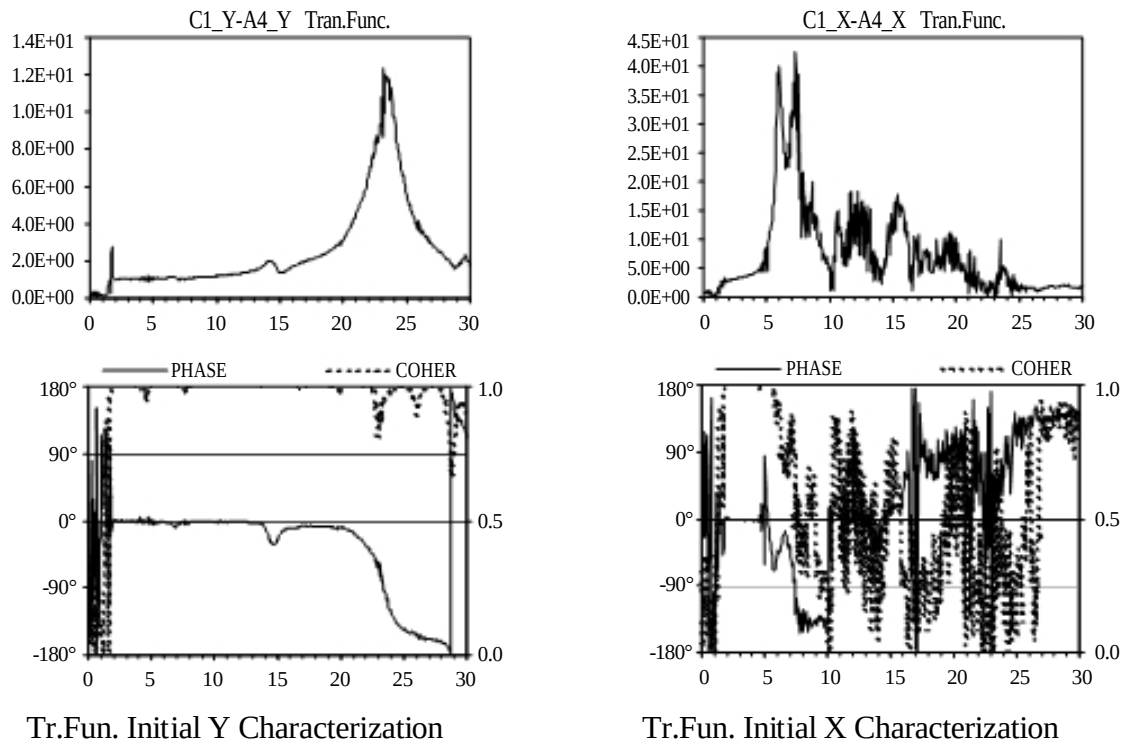


Fig. 19